

1. (20 marks) For $(a, b, k) \in \mathbb{R}^3$, consider the following quadratic form

$$Q(\mathbf{x}) = ax_1^2 + kx_2^2 + bx_3^2 + 2ax_1x_2 + 2bx_2x_3.$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A .
- (b) (12 marks) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all parameter values for a , b , and k such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*?
- (c) (4 marks) For $b = a$, plot the pairs (a, k) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$.
2. (20 marks) Consider two *quasi-concave* functions $f : U \rightarrow \mathbb{R}$ and $g : U \rightarrow \mathbb{R}$ defined on some convex set U in $\mathbb{R}^n$.

- (a) (15 marks) Show that their sum,

$$S(\mathbf{x}) = f(\mathbf{x}) + g(\mathbf{x}) \text{ for all } \mathbf{x} \in U,$$

is not necessarily *quasi-concave* function. Then, establish that the sum S may even be *quasi-convex*.

- (b) (5 marks) What are the implications of part (a) for an optimization problem with a *quasi-concave* the objective function and *quasi-convex* constraints? Discuss briefly.
3. (40 marks) For $a > 0$, consider the problem:

$$\max_{x,y} 3a \log(y+1) + x \quad \text{s.t.} \quad y + ax \leq 1, \quad y \geq 0, \quad x \geq a.$$

- (a) (6 marks) Is the objective function concave? Are the constraints convex? For what values of $a > 0$ is Slater condition satisfied? So, for what values of $a > 0$, can you solve this optimization problem using the *Kuhn-Tucker method*?
- (b) (6 marks) Suppose the restrictions on a found in part (a) hold. Write down the *Kuhn-Tucker necessary first order conditions* that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?
- (c) (20 marks) Suppose the restrictions on a found in part (a) hold. Solve the constrained optimization problem in terms of a .
- (d) (8 marks) Suppose the restrictions on a found in part (a) hold. Use the Envelope Theorem to find the slope of value function $v(a)$ with respect to the parameter a . Is there a value for a that maximizes the value function $v(a)$ of this problem?
4. (20 Marks) Consider the correspondence $\mathbf{h} : [0, 1]^2 \rightrightarrows [0, 1]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} 1 & \text{if } x_2 \in (3/4, 1] \\ 2x_2 - 1/2 & \text{if } x_2 \in [1/4, 3/4] \\ 0 & \text{if } x_2 \in [0, 1/4) \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [0, a] & \text{if } x_1 = 1 \\ a & \text{if } x_1 \in (0, 1) \\ [0, a] & \text{if } x_1 = 0 \end{cases},$$

for a scalar $a \in [0, 1]$. Briefly, argue why *Kakutani fixed point theorem* applies. Then, for all values of $a \in [0, 1]$ find all the *fixed points* of this correspondence.